

THE UNIVERSITY OF HONG KONG
FACULTY OF BUSINESS AND ECONOMICS

MSBA7033 Generative Artificial Intelligence

GENERAL INFORMATION

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Course Website: Moodle

Required Textbook: Chen, W., & Chen, L. (2025). *Generative AI for Business: Framework, Techniques, and Governance* (1st ed.). GenAI Flows Publishing. ISBN-13: 979-8999716125. <https://genai4all.org/book>

Recommended Book: The Founder's Playbook: Building an AI-Native Startup <https://claude.com/blog/the-founders-playbook>

Programming Prerequisites: None. No prior technical background or programming language is required.

Recommended Reading Articles:

1. Bengio, Y., Ducharme, R., Vincent, P., & Jauvin, C. (2003). *A neural probabilistic language model*. Journal of Machine Learning Research, 3(Feb), 1137-1155.
2. Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013). *Efficient estimation of word representations in vector space*. arXiv preprint arXiv:1301.3781.
3. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, N., Kaiser, Ł., & Polosukhin, I. (2017). *Attention is all you need*. Advances in Neural Information Processing Systems, 30. <https://arxiv.org/abs/1706.03762>
4. Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., ... & Amodei, D. (2020). *Language models are few-shot learners*. Advances in neural information processing systems, 33, 1877-1901.
5. Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). *On the dangers of stochastic parrots: Can language models be too big?* Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT '21), 610–623. <https://doi.org/10.1145/3442188.3445922>
6. Achiam, J., Adler, S., Agarwal, S., Ahmad, L., Akkaya, I., Aleman, F. L., ... & McGrew, B. (2023). *Gpt-4 technical report*. arXiv preprint arXiv:2303.08774.
7. Eloundou, T., Manning, S., Mishkin, P., & Rock, D. (2024). *GPTs are GPTs: Labor market impact potential of LLMs*. Science, 384(6702), 1306-1308.
8. Liu, A., Feng, B., Xue, B., Wang, B., Wu, B., Lu, C., ... & Piao, Y. (2024). *Deepseek-v3 technical report*. arXiv preprint arXiv:2412.19437.
9. Esser, P., Kulal, S., Blattmann, A., Entezari, R., Müller, J., Saini, H., ... & Rombach, R. (2024, July). *Scaling rectified flow transformers for high-resolution image synthesis*. In Forty-first international conference on machine learning.
10. Chatterji, A., Cunningham, T., Deming, D. J., Hitzig, Z., Ong, C., Shan, C. Y., & Wadman, K. (2025). *How people use ChatGPT* (No. w34255). National Bureau of Economic Research.

COURSE DESCRIPTION

Generative AI for Business is designed to equip students with the knowledge and skills needed to leverage generative AI and large language models (LLMs) in business contexts.

The course examines the core mechanisms of generative AI, with a focus on LLMs. Students explore advanced techniques in prompt engineering, retrieval-augmented generation (RAG), custom GPT applications, AI agents, and agentic workflows.

The curriculum balances theoretical knowledge with hands-on exercises, enabling students to apply generative AI innovations to real-world business challenges. It also develops students' ability to assess governance, ethics, opportunities, and risks associated with generative AI.

COURSE OBJECTIVES

- CO1: Explain the foundational concepts, architecture, and training methods of generative AI and LLMs.
- CO2: Apply prompt-engineering principles and advanced techniques to business problems.
- CO3: Integrate retrieval-augmented generation into LLM applications.
- CO4: Develop custom GPT applications, AI agents, and agentic workflows.
- CO5: Evaluate business use cases and the future-of-work implications of generative AI.
- CO6: Discuss the governance and ethical opportunities and challenges introduced by generative AI.

PROGRAMME LEARNING OUTCOMES

- PLO1: Acquisition of knowledge of the discipline
- PLO2: Application and integration of knowledge
- PLO3: Inculcating professionalism and leadership
- PLO4: Developing global outlook
- PLO5: Mastering communication skills

COURSE LEARNING OUTCOMES

Course Learning Outcomes

Aligned Programme Learning Outcomes

- CLO1: Articulate the foundational concepts of generative AI.
- CLO2: Implement effective prompt-engineering strategies and build robust LLM applications and custom chatbots for business needs.
- CLO3: Integrate RAG into LLM applications to provide contextually relevant responses.
- CLO4: Develop custom GPT applications and tools.
- CLO5: Design and implement AI agents and agentic workflows to automate business processes.
- CLO6: Engage in thoughtful discussion of generative AI governance, opportunities, and challenges.

PLO1, PLO2
 PLO1, PLO2, PLO5
 PLO1, PLO2
 PLO1, PLO2
 PLO2, PLO3, PLO5
 PLO3, PLO4, PLO5

COURSE TEACHING AND LEARNING ACTIVITIES

Course Teaching and Learning Activities	Contact Duration	Study Load in Percentage of Study
T&L1. Interactive lectures and demonstrations	20 hours	17%
T&L2. Hands-on exercises and in-class discussions	10 hours	8%
T&L3. Discussion boards, homework, and exams	60 hours	50%

T&L4. Lecture preparation and self-study	30 hours	25%
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COURSE ASSESSMENTS

Assessment Methods	Assessment Period or Due Date	Weight in Percentage	Aligned Course Learning Outcomes
Discussion Board	Sections 1-8	10%	CLO1-6
Attendance	Sections 1-10	10%	CLO1-6
Homework 1	Section 2	10%	CLO1, 2
Homework 2	Section 3	10%	CLO1, 3
Homework 3	Section 4	10%	CLO1, 4, 5
Homework 4	Section 7	10%	CLO6
Mid-term Examination	Section 5	20%	CLO1-5
Final Examination	Section 10	20%	CLO1-6

COURSE CONTENT AND TEACHING SCHEDULE

Section 1: Course Introduction and Foundational Concepts of Generative AI

- Fundamentals of generative AI and LLMs
- Transformer architecture and LLM training methods
- Business cases of LLMs and useful tools
- Readings: Chapters 1-2 | Deliverable: Discussion Board 1

Section 2: Prompt Engineering

- Principles, advanced techniques, and hands-on practice
- Readings: Chapters 3-4 | Deliverables: Homework 1; Discussion Board 2

Section 3: Retrieval-Augmented Generation (RAG)

- RAG pipeline and practice; text embeddings and vector databases
- Readings: Chapters 5-6 | Deliverables: Homework 2; Discussion Board 3

Section 4: AI Agents

- Agentic reasoning patterns and hands-on practice
- Reading: Chapter 7 | Deliverables: Homework 3; Discussion Board 4

Section 5: Mid-term Examination

Section 6: Agentic Workflow

- User-defined and autonomous workflows; hands-on practice
- Reading: Chapter 8 | Deliverable: Discussion Board 5

Section 7: Governance and Ethics of Generative AI

- Readings: Chapters 9-10 | Deliverables: Homework 4; Discussion Board 6

Section 8: Generative AI and the Future of Work

- Readings: Chapters 11-12 | Deliverable: Discussion Board 7

Section 9: Research Frontier - The Case of Emotion AI

Section 10: Final Examination

PROCESSES FOR STUDENT FEEDBACK ON COURSE

Conduct pre-term, mid-term, and final surveys using online forms and respond to student feedback.

COURSE POLICY

An orderly learning environment is essential. Disruptive behaviour is unacceptable. Academic dishonesty includes cheating, plagiarism, unauthorized collaboration, falsifying academic records, and any act designed to avoid honest participation in the learning process. Use of generative AI in assignments or reports is permitted only when expressly authorized by the relevant assessment instructions. Academic dishonesty may result in a failing grade and further action under University regulations.

ADDITIONAL COURSE INFORMATION

Assignments and discussion-board submissions will be made through Moodle. Late work will not be accepted unless prior approval or an applicable University policy applies. Lecture notes and self-learning materials will be uploaded to Moodle. The course schedule is subject to change.

STANDARDS FOR ASSESSMENT

Course Grade Descriptors

A+, A, A-	Demonstrate a strong understanding of all relevant knowledge Handling questions professionally High participation in discussions Present arguments that have an element of originality Excellent performance in assignments
B+, B, B-	Demonstrate a good understanding of all relevant knowledge Handling questions in a logical way Good participation in discussions Present arguments that go beyond the lecture and textbook Good performance in assignments
C+, C, C-	Demonstrate a basic understanding of the concepts involved Fairly address questions as set Some participation in discussions Present arguments in a well-structure manner Acceptable performance in assignments
D+, D	Demonstrate a minimum understanding of the concepts involved Barely address questions as set Minimal or no participation in discussions Present arguments in a marginally acceptable manner Marginally acceptable performance in assignments
F	Demonstrate a poor understanding of the concepts involved Unable or unwilling to handle questions Minimal or no participation in discussions Present arguments poorly Poorly performance in assignments

Assessment Rubrics for Participation

A+, A, A-	High participation in discussions Demonstrate a strong understanding of all relevant knowledge Handling questions professionally Present arguments that have an element of originality
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	Respect others and follow the class rules (no chatting and do not use cell phone)
B+, B, B-	Good participation in discussions Demonstrate a good understanding of all relevant knowledge Handling questions in a logical way Present arguments that go beyond the lecture and textbook Respect others and follow the class rules (no chatting and do not use cell phone)
C+, C, C-	Some participation in discussions Demonstrate a basic understanding of the concepts involved Fairly address questions as set Present arguments in a well-structure manner Respect others and follow the class rules (no chatting and do not use cell phone)
D+, D	Minimal or no participation in discussions Demonstrate a minimum understanding of the concepts involved Barely address questions as set Present arguments in a marginally acceptable manner Respect others and follow the class rules (no chatting and do not use cell phone)
F	Minimal or no participation in discussions